# 崑山科技大學日間部資訊工程系四技人工智慧應用組

# 課程科目表

## Kun Shan University Day Program Department of Information Engineering - AI Application Division

## Four-Year Technical Program Course Schedule

課程名稱(講授時數－實習時數－學分數)
Course Title (Lecture Hours – Lab Hours – Credits)

111.11.10 系教學與課程規劃委員會議通過
111.11.14 院課程委員會議通過
112.02.13 校課程委員會議通過
110 學年度入學適用
Effective for students admitted in Fall, 2021

| 第一學年<br>1st Academic Year | | 第二學年<br>2nd Academic Year | | 第三學年<br>3rd Academic Year | | 第四學年<br>4th Academic Year | |
|---|---|---|---|---|---|---|---|
| 第一學期<br>1st Semester | 第二學期<br>2nd Semester | 第一學期<br>1st Semester | 第二學期<br>2nd Semester | 第一學期<br>1st Semester | 第二學期<br>2nd Semester | 第一學期<br>1st Semester | 第二學期<br>2nd Semester |
| **共同必修科目(合計 30 學分) University-wide Required Courses (30 Credits)** | | | | | | | |
| 體育分組(一)<br>Physical Education (I)<br>(2-0-2)<br>中文閱讀與表達（一）<br>Chinese Reading and Expression (I)<br>(2-0-2)<br>英文聽力與閱讀(一)<br>English Listening and Reading (I)<br>(2-0-2) | 體育分組(二)<br>Physical Education (II)<br>(2-0-2)<br>勞作教育(一)<br>Labor Education(I)<br>(0-1-0.5)<br>中文閱讀與表達（二）<br>Chinese Reading and Expression (II)<br>(2-0-2)<br>英文聽力與閱讀(二)<br>English Listening and Reading (II)<br>(2-0-2)<br>校訂<br>分類通識<br>School-designated Liberal Education<br>(2-0-2) | 勞作教育(二)<br>Labor Education(II)<br>(0-1-0.5)<br>校訂<br>分類通識<br>School-designated Liberal Education<br>(2-0-2)<br>院訂<br>核心通識<br>**科技英文**<br>College-designated General Core<br>**Scientific and Technological English**<br>(2-0-2)<br>院訂<br>分類通識<br>College-designated Liberal Education<br>(2-0-2) | 服務學習與公民教育<br>Public Service and Civic Education<br>(0-1-0)<br>校訂<br>分類通識<br>School-designated Liberal Education<br>(2-0-2) | 生涯發展與職業倫理<br>Career Development and Work Ethics<br>(2-0-2)<br>院訂<br>分類通識<br>College-designated Liberal Education<br>(2-0-2) | 校訂核心通識<br>--外語選讀***<br>School-designated General Core ***<br>(2-0-2) | | *英文能力檢定<br>English Proficiency Examination<br>(1-0-1) |
| 6-0-6 | 8-1-8.5 | 6-1-6.5 | 2-1-2 | 4-0-4 | 2-0-2 | | 1-0-1 |
| **共同選修科目(合計 1 學分) University-wide Elective Courses (1 Credit)** | | | | | | | |
| | | 外語選讀<br>(第二外語)***<br>Foreign Language second language<br>(2-0-2) | 外語選讀<br>(第二外語)***<br>Foreign Language second language<br>(2-0-2) | 外語選讀(職場英文) ***<br>Foreign Language-Workplace English<br>(2-0-2) | | 跨領域學習護照**<br>Interdisciplinary Learning Passport **<br>(1-0-1) | |
| | | | | | | 1-0-1 | |

註*：*「**英文能力檢定**」依「崑山科技大學學生英文基本能力檢定實施辦法」。
"English Proficiency Examination" is enacted according to "Kun Shan University Guidelines for English Proficiency Requirement."
**跨領域學習護照，依「崑山科技大學通識學習護照實施辦法」辦理。
"Interdisciplinary Learning Passport" is enacted according to "Kung Shan University Guidelines for General Education Learning Passport."
***「外語選讀」為選項式必修科目，課程包括「第二外語」及「職場英文」，擇一門修習即可。
"Foreign Languages" compromises "Foreign Languages (other than English)" and "Workplace English", among which students choose 1 course.

**院訂必修科目（合計 8 學分）College-Designated Required Courses (Total: 8 Credits)**

| 第一學年<br>1st Academic Year | | 第二學年<br>2nd Academic Year | | 第三學年<br>3rd Academic Year | | 第四學年<br>4th Academic Year | | |
|---|---|---|---|---|---|---|---|---|
| 第一學期<br>1st Semester | 第二學期<br>2nd Semester | 第一學期<br>1st Semester | 第二學期<br>2nd Semester | 第一學期<br>1st Semester | 第二學期<br>2nd Semester | 第一學期<br>1st Semester | 第二學期<br>2nd Semester | |
| 微積分<br>Calculus<br>(3-0-3)<br>工程技術概論<br>Introduction to Engineering Technology<br>(2-0-2) | 工程數學<br>Engineering Mathematics<br>(3-0-3) | | | | | | | 計算機基礎理論<br>Fundamental Theory for Computing |
| 5-0-5 | 3-0-3 | | | | | | | |

## 專業必修科目(合計 54 學分) Professional Required Courses (Total: 54Credits)

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 程式設計(一)<br>Programming Design (I)<br>（1-2-3） | 程式設計(二)<br>Programming Design (II)<br>（1-2-3） | 資料結構<br>Data Structure<br>(3-0-3) | | | 離散數學<br>Discrete Mathematics<br>(3-0-3) | | | 程式設計<br>Programming Design |
| | | Python 人工智慧實務<br>Artificial Intelligence Practice in Python<br>(1-2-3) | | 軟體工程<br>Software Engineering<br>(3-0-3) | | | | 軟體開發<br>Software Development |
| | | *Linux 作業系統實務<br>Practice of Linux Operating System<br>(1-2-3) | 計算機網路實務<br>Computer Network Practice<br>(1-2-3) | 雲端運算概論<br>Introduction to Cloud Computing<br>(3-0-3) | | | | 雲端運算技術<br>Cloud Computing Technology |
| | 人工智慧概論<br>Introduction to Artificial Intelligence<br>(3-0-3) | 資料庫實務<br>practice of database<br>(1-2-3) | 機器學習<br>Machine Learning<br>(1-2-3) | 深度學習實務<br>Deep Learning Practice<br>(1-2-3) | 大數據分析<br>Big Data Analysis<br>(1-2-3) | | | 人工智慧應用<br>Applications of Artificial Intelligence |
| | | | | | 創新智慧生活科技<br>Innovative Smart Life Technology<br>(2-1-3) | 學術倫理*<br>Academic Ethics<br>(1-0-1) | | 創新實務<br>Innovative Practice |
| | | | | | *專題製作（一）<br>Special Project (I)<br>（0-2-2） | *專題製作（二）<br>（0-2-2）<br>Special Project (II)<br>暑期校外實習<br>Summer School Internship<br>(0-*-4) | 專業認證<br>Professional certification<br>(0-1-0) | 整合性課程<br>Integrated Courses |
| 1-2-3 | 4-2-6 | 6-6-12 | 2-4-6 | 7-2-9 | 6-5-11 | 1-2-7 | 0-1-0 | |

註*：「學術倫理」之授課與學分取得依課務組網頁之「學術倫理教育實施辦法」辦理。

Note*：The implementation of 'Academic Ethics' and the earning of credits are conducted in accordance with the 'Guidelines for Academic Ethics Education' published on the Curriculum Division's website.

## 專業選修科目(依每學期實際情況開課，含專業養成課程 8 學分) Professional Elective Courses (Offered based on the actual circumstances of each semester, including 8 credits of professional development courses)

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|---|---|---|---|---|---|---|---|---|
| | | 影像處理概論<br>Introduction to Image Processing<br>(3-0-3) | 演算法<br>Algorithm<br>(3-0-3) | 機率與統計<br>Probabilities and Statistics<br>(3-0-3) | | 專案管理<br>Project Management<br>(3-0-3) | | 計算機數學與理論<br>Mathematics for Computing and Theory |

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| | | | 伺服網頁程式設計<br>Server Web Design<br>(1-2-3) | | | | | 軟體開發<br>Software Development |
| | | | 微算機原理與實習<br>Microcomputer principle and practices<br>(1-2-3) | | | | | 嵌入式基礎<br>Embedded Fundamental |
| | 物聯網於綠能之應用<br>IOT application in Green Energy<br>(1-2-3) | | 伺服器架設實務<br>Server Construction Practice<br>(1-2-3) | 智慧能源與物聯網<br>Intelligent Energy and IOT<br>(1-2-3) | | 物聯網<br>Internet of Things<br>(1-2-3) | 多媒體智慧聯網實務<br>Multimedia Smart Networking Practice<br>(2-1-3) | 智慧聯網<br>Smart Network |
| 資訊工程導論<br>Outline of Information Technology Concepts<br>(3-0-3) | HTML 5 程式設計<br>HTML 5 Programming<br>(1-2-3) | | | | 自然語言處理與聊天機器人實務<br>Practice of Natural Language Processing and Chatbot<br>(3-0-3) | 電子商務實務<br>Practice of E-Commerce<br>(3-0-3) | 雲端服務導向架構中介軟體<br>Cloud Service-Oriented Architecture Middleware<br>(3-0-3) | 智慧服務<br>Intellegient Service |
| | 網路攻防實務<br>Cyber Offense and Defense Practice<br>(1-0-1)<br>CTF 搶旗大賽實務技術<br>CTF Flag-Raising Contest Practical Technology<br>(1-0-1) | 網站安全<br>Web Security<br>(2-1-3) | 系統安全<br>System security<br>(3-0-3) | 雲端安全<br>Cloud Security<br>(3-0-3) | | | 安全實務<br>Safety practice<br>(2-1-3)<br>物聯網與資訊安全<br>Internet of Things and Information Security<br>(2-1-3) | 資訊安全<br>Information Security |
| | | 類神經網路<br>Neural Network<br>(1-2-3) | 模糊系統<br>Fuzzy System<br>(1-2-3) | 電腦視覺實務<br>Computer Vision Practice<br>(1-2-3)<br>演化式演算法<br>Evolutionary Algorithms<br>(1-2-3) | 影像處理<br>Image Processing<br>(3-0-3)<br>AOI 檢測技術實務<br>AOI testing technology practice<br>(1-2-3)<br>機器學習<br>Machine Learning<br>(3-0-3) | 智慧型機器人系統應用專題<br>Special Study on Intelligent Robot System Applications<br>(1-2-3)<br>AIoT 應用實務<br>AIoT Applications and Practices<br>(1-2-3)<br>人工智慧<br>Artificial Intelligence<br>(3-0-3)<br>**深度學習實務**<br>Deep Learning Practice<br>(3-0-3) | 智慧聯網自動光學檢測實務專題<br>Smart Networked AOI Special Project<br>(1-2-3)<br>Python 金融大數據分析<br>Financial Bigdata Analysis with Python<br>(1-2-3)<br>人工智慧實務專題<br>Practical Projects in Artificial Intelligence<br>(1-2-3)<br>跨平台程式資料庫應用<br>Applications for Cross-Platform Programming and Database<br>(1-2-3) | 智慧應用<br>Intelligent Application |

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|---|---|---|---|---|---|---|---|---|
| 第一學期<br>1st Semester | 第二學期<br>2nd Semester | 第一學期<br>1st Semester | 第二學期<br>2nd Semester | 第一學期<br>1st Semester | 第二學期<br>2nd Semester | 第一學期<br>1st Semester | 第二學期<br>2nd Semester | |
| | | Swift IOS app 開發<br>Swift iOS Applications Development<br>(3-0-3)<br>App 程式設計<br>App Programming<br>(1-2-3) | 行動裝置 App 專題製作<br>Mobile Device App Project Creation<br>(3-0-3) | iOS 應用開發<br>iOS app Development<br>(3-0-3) | 行動裝置遊戲開發<br>Mobile Device Game Development<br>(3-0-3) | 行動 APP 專案開發<br>Mobile APP Project Development<br>(3-0-3) | | App 開發<br>App Development |
| 跨網頁平台設計<br>Cross-web platform design<br>(3-0-3) | 網站開發與安全<br>web development and application security<br>(3-0-3) | | | | D3.js 程式設計<br>D3.js programming<br>(3-0-3) | JSP 程式設計 JSP programming<br>(3-0-3) | Ajax 程式設計<br>Ajax Programming<br>(3-0-3) | 網站程式設計<br>Web Programming |
| python 程式設計<br>python Programming<br>（3-0-3） | C#程式設計<br>C# programming<br>(1-1-2) | | | | | | Design Pattern 實務<br>Design Pattern Practice<br>(1-1-2) | 程式設計<br>Programming Design |
| 視覺設計實務<br>Visual Design Practice<br>(3-0-3) | 網頁視覺設計實務<br>Web Visual Design Practice<br>(3-0-3) | | | 手機 UI 設計<br>Mobile UI Design<br>(3-0-3) | | 資料視覺化<br>Data Visualization<br>(3-0-3) | | 互動設計<br>Interactive Design |
| | | | | 專業校外實習(一)<br>Industrial Internship(I)<br>(0-2-2) | 專業校外實習(二)<br>Industrial Internship(II)<br>(0-2-2) | 學期校外實習(一)<br>Semester Internship (I)<br>(0-*-9)<br>學年校外實習(一)<br>Annual Internship (I)<br>(0-*-9) | 學期校外實習(二)<br>Semester Internship (II)<br>(0-*-9)<br>學年校外實習(二)<br>Annual Internship (II)<br>(0-*-9) | 企業實習<br>Industrial Internship |
| 微服務伺服器實務<br>Microservice server practice<br>(0-1-1)<br>專案開發與應用實務<br>Project Developement and Application Practice<br>(0-1-1)<br>關聯式資料庫實務<br>Relational Database Practice<br>(0-1-1) | GTM 實務<br>Google Tag Manager Practice<br>(0-1-1)<br>軟體測試實務<br>Software Testing Practicalities<br>(0-1-1)<br>人機介面設計專題<br>Human Machine Interface Design Project<br>(0-1-1) | 資電整合應用<br>Applications of Information and electronic integration<br>(0-1-1)<br>跨平台程式設計專題<br>Cross-platform programming languages<br>(0-1-1)<br>網頁前後端整合實務<br>Web Integration Practices on Front-End and Back-End<br>(0-1-1) | M2M 聯網實務<br>Practice of Machine to Machine<br>(0-1-1)<br>爬蟲程式應用實務<br>Application of python in data crawling<br>(0-1-1)<br>物聯網巨量資料實務<br>IOT BigData Practice<br>(0-1-1) | 微電腦視覺實務<br>Microcomputer Vision and Practice<br>(0-1-1)<br>網路串流技術應用<br>The Application of Network Streaming Technology<br>(0-1-1)<br>機器學習實務<br>Machine Learning Practice<br>(0-1-1) | 微控制器資訊感測<br>The sensing of microcontroller<br>(0-1-1)<br>行動影音技術專題<br>Audiovisual Technology in Mobile Devices<br>(0-1-1)<br>iOS 行動應用實務<br>iOS Mobile App Practices<br>(0-1-1) | 機器手臂應用<br>Robot Arm Applications<br>(0-1-1)<br>CTF 資安實戰<br>CTF and Security<br>(0-1-1)<br>AI 生醫感測實務<br>AI Biomedical Sensor and Measurement<br>(0-1-1) | 感測器應用實務<br>Sensor Applications<br>(0-1-1)<br>機器學習與資訊安全實務<br>Machine learning and Information Security Practice<br>(0-1-1)<br>物聯網應用實務<br>IoT Appilication Practice<br>(0-1-1) | 專業養成課程<br>Professional Development Courses |

跨域選修科目(合計 9 學分) Interdisciplinary Elective Courses (Total: 9 Credits)

*註：可選修外系專業選修，但不可為學術倫理、專題製作、專題研究、創新智慧生活科技、校外實習等課程。

*Note：Students may take professional electives from other departments, but the courses must not include Academic Ethics, Project Production, Research Projects, Innovative Smart Living Technology, or Off-Campus Internships.

最低畢業學分：130 學分 Minimum for Total Credits: 130 Credits

註：

1. **本系人工智慧應用組畢業學分為 130 學分，必修 92 學分、選修 38 學分、跨域選修 9 學分。**
   The total graduation requirement for this department is **130 credits**, including **92required credits**, **38 elective credits**, and **9 interdisciplinary elective credits**.
2. 選修科目，本系得視實際需求修訂之。
   The elective courses may be revised by the department as needed.
3. 專業必修課程第一次修習於本系修課，再次修習時須事先徵得系之同意後可在他系修習同性質之課程。
   Professional required courses must be taken within this department for the first attempt. If retaken, approval from the department is required before enrolling in an equivalent course offered by another department.
4. **學生須修畢本校所開設之任一跨院系學分學程，始得畢業。**
   Students must complete at least one inter-college or interdisciplinary credit program offered by the university in order to graduate.
5. 上列**專業選修**科目，自行網路選課，低年級除特殊情況經系主任同意者外不得修習高年級課程。
   The above professional elective courses should be selected online by the students themselves. Lower-year students are not allowed to take upper-year courses unless special permission is granted by the department chair.
6. **學生除選修本系專業選修課程外，可多選修產業學程、跨院系跨領域學程及課程等，以符合就業市場之需求，增加就業力。**
   In addition to taking the department's professional elective courses, students are encouraged to take additional industry-oriented, interdisciplinary, or cross-college programs and courses to meet job market demands and enhance their employability
7. 暑期校外實習之實施為至少 8 週 320 小時，另外訂定執行辦法。學期校外實習之實施為開設 9 學分以上，至少為期 4.5 個 月之校外實習課程，修讀實習課程期間，除依各校訂定定期返校之座談會或研習活動等外，學生應全職於實習機構實習。
   Summer Off-campus Internship shall be implemented for a minimum of 8 weeks and 320 hours; the implementation regulations shall be established separately. Semester Off-campus Internship shall be offered as a course with 9 or more credits and a minimum duration of 4.5 months. During the internship period, students shall intern full-time at the host organization, with the exception of attending scheduled seminars or workshops organized by the university.
8. **系專業必修科目「專業認證」，需於在本校就讀期間取得本系所認定之乙級以上（含）專業證照 1 張、取得參加全國性資訊相關競賽證明乙份、通過系專題展及繳交專題報告，始得通過。**
   For the departmental required course **"Professional Certification"**, students must, during their period of study at the university, obtain **at least one professional certificate at Class B or higher** as recognized by the department, **one proof of participation in a national information-related competition**, and **successfully complete the departmental project exhibition and submit a project report** in order to pass the course.